

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

The trust gap in commercial LLM APIs: providers may covertly swap advertised models for cheaper substitutes.

Source: arxiv.org/abs/2504.04715

ML Researchers • AI Security Practitioners • Cloud Infrastructure Engineers

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Core Problem

Commercial LLM API providers charge for specific models (e.g., Llama-3-70B, Qwen2-72B), but the black-box nature of APIs means users cannot verify the advertised model is actually being served.

Providers face strong economic incentives to covertly substitute cheaper alternatives — quantized versions, smaller models, or entirely different architectures — to reduce GPU costs while maintaining pricing.

Formal Framework

H_0 : API samples from $P_{\text{spec}}(y|x)$

H_1 : API samples from $P_{\text{actual}}(y|x)$

1

Smaller Model

Serve an 8B parameter model instead of the advertised 70B.

2

Quantized Variant

Replace FP16 weights with FP8 or INT8 quantized versions.

3

Fine-Tuned / Updated

Substitute a modified or updated checkpoint without disclosure.

4

Different Family

Serve an entirely different model architecture.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Insight: For most models, FP8 scores fall within the error bars of the original, especially on MMLU and GSM8K. The substitution signal is nearly invisible.

Higher Temperature Amplifies Performance Gaps but Increases Noise

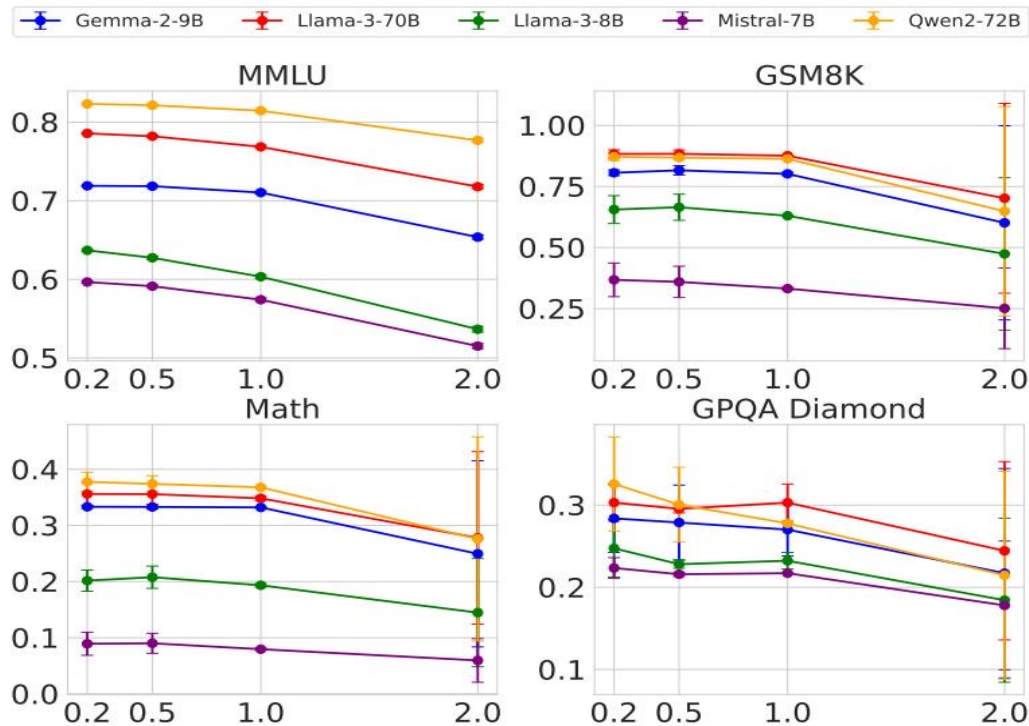


Figure 1: Benchmark accuracy vs. temperature for five models across four tasks

Temperature Scaling Experiments

Experiments tested temperatures 0.2, 0.5, 1.0, and 2.0 across all five model families.

Key Observation:

At higher temperatures (especially 2.0), performance degrades across all models. The gap between original and substitute narrows or becomes noisy.

This further obscures the substitution signal, making statistical detection even more challenging under realistic sampling conditions.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: Log probability variation across inference frameworks and hardware for Llama-3-8B

Production Inference Variability

The same Llama-3-8B model produces different token-level log probabilities depending on:

- Software framework: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- GPU hardware: A100 vs H100

Consequence:

An auditor cannot distinguish between a legitimate model on different infrastructure and an actual substitution. Log-probability fingerprinting is unreliable in practice.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Randomized Substitution Attack

Provider routes each query to the cheaper substitute with probability p , and to the genuine model with probability $(1-p)$.

Mixed distribution: $p \cdot P_{\text{actual}} + (1-p) \cdot P_{\text{spec}}$

Convergence Property

As $p \rightarrow 0$, the mixed distribution converges to P_{spec} . The auditor needs exponentially more samples to detect the substitution, making software-only detection fundamentally query-prohibitive.

Randomized Substitution

Route only fraction p of queries to cheaper model.
Mixed distribution approaches genuine as p decreases.

Benchmark Evasion

Detect known benchmark prompts and route them to genuine model. Serve substitutes for all regular traffic.

These adversarial strategies make software-only detection fundamentally query-prohibitive.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Require impractically large query budgets; fail against subtle or randomized substitutions.	Query volume grows exponentially as substitution rate p decreases. Cost exceeds API savings.
Log-Probability Fingerprinting	Defeated by production-level inference nondeterminism across software versions and GPU hardware.	Cannot distinguish legitimate infrastructure differences from actual model substitution.
Adversarial Countermeasures	Randomized routing and benchmark evasion further undermine all detection approaches.	Provider controls routing logic; auditor has no visibility into backend dispatch decisions.

No software-only method provides reliable, cost-effective detection of covert model substitution in production LLM APIs.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

What Are TEEs?

Hardware-secured enclaves that provide cryptographic attestation of the code and data being executed.

The provider can cryptographically prove:

- Specific model weights are loaded
- Correct inference code is running
- Execution occurs inside the attested enclave

TEEs vs. Software-Only Methods

Provable

Cryptographic attestation, not statistical inference

Efficient

Only modest performance overhead vs. ZKPs

Robust

Immune to randomized routing and evasion

Practical Deployment Path

NVIDIA Confidential Computing on H100 GPUs provides a deployable TEE solution for LLM inference today.

Unlike Zero-Knowledge Proofs (ZKPs), which are computationally prohibitive for large models, TEEs are practical and deployable now.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and practically undetectable via software-only methods.

2

FP8 quantization barely affects benchmark scores, making it an attractive but nearly invisible substitute.

3

Inference nondeterminism across software versions and GPU hardware defeats log-probability fingerprinting.

4

Randomized substitution and benchmark evasion make statistical detection query-prohibitive.

5

Trusted Execution Environments are the only currently viable path to verifiable LLM APIs.

Call to Action: The research community should prioritize TEE-based attestation standards for LLM API integrity.